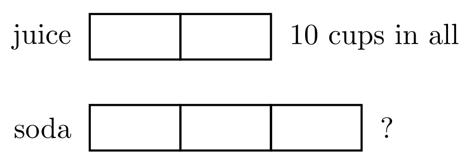
Tape Diagrams and Double Number Lines for Ratios

Reading and completing tape diagrams, filling in double number lines, finding a unit rate from the line, and comparing two ratios with the diagrams

Two pictures, one idea. A **tape diagram** shows a ratio as equal parts: every box is worth the same amount, so finding the value of *one box* unlocks the whole problem. A **double number line** pairs two quantities that grow together: the two lines are marked so that matching points sit above each other.

- On a tape diagram: value of one box = known amount ÷ number of boxes it covers.
- On a double number line: multiply or divide *both* lines by the same number to move to a new pair.
- The **unit rate** is the number that sits above 1 on the other line. It makes every later question one multiplication.
- Label your diagram before you compute, and write the units in your answer.

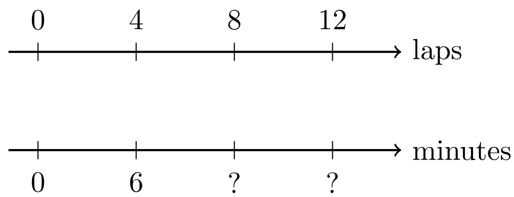
1. A punch recipe mixes juice and soda in the ratio 2 : 3. The tape diagram is drawn with 10 cups of juice. How many cups of soda are needed?



First write the value of one box, then the amount of soda.

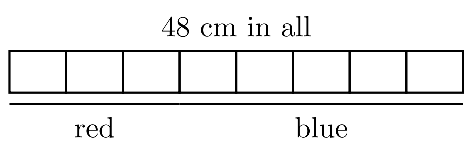
Answer: ________________ cups

2. A swimmer keeps a steady pace: 4 laps takes 6 minutes. Fill in the two missing labels on the double number line, then find how long 10 laps takes.



Answer: ________________ minutes

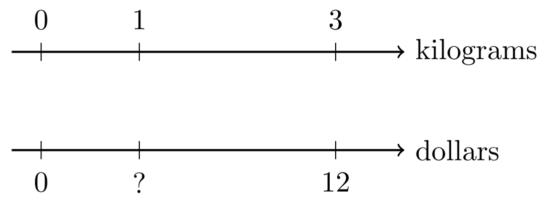
3. A 48 centimetre ribbon is cut into red and blue pieces in the ratio 3 : 5. Each box below is one equal part.



Find the value of one box, then the length of the red piece and the length of the blue piece.

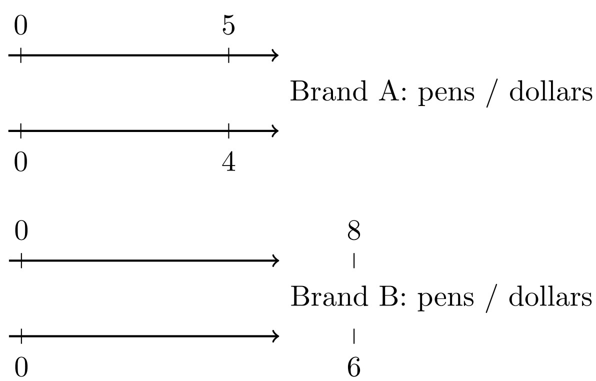
Answer: ________________ cm

4. Apples cost the same for every kilogram: 3 kilograms cost 12 dollars. Use the double number line to find the cost of 1 kilogram, then the cost of 7 kilograms, both in dollars.



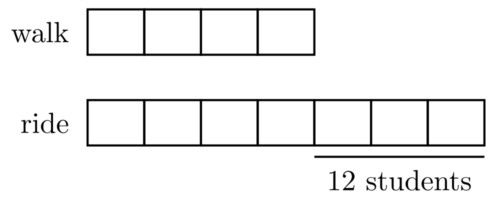
Answer: ________________

5. Two shops sell the same pens. Brand A: 5 pens for 4 dollars. Brand B: 8 pens for 6 dollars. Find the cost of one pen for each brand, then write $>$ or $<$ between the two costs and name the cheaper brand.



Answer: ________________

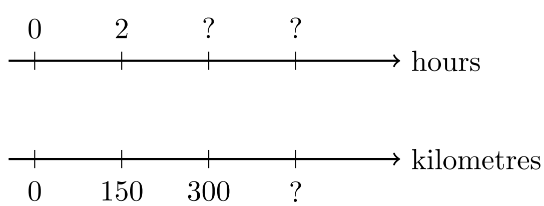
6. In a class the ratio of students who walk to school to students who ride is 4 : 7. There are 12 more riders than walkers.



Let x be the number of students one box stands for. Write and solve an equation for x, then find how many students walk and how many ride.

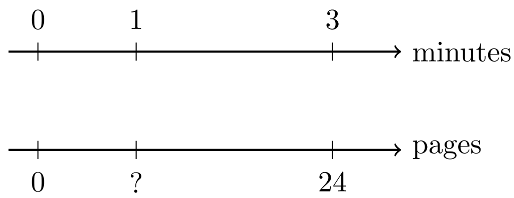
Answer: ________________

7. A car travels at a steady speed: 150 kilometres in 2 hours. Use the double number line. **(a)** How far does it travel in 5 hours? **(b)** How long does 300 kilometres take, in hours?



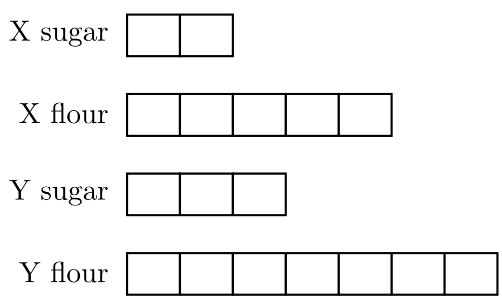
Answer: ________________

8. A printer prints 24 pages in 3 minutes at a steady rate. **(a)** Fill in the pages that print in 1 minute — the unit rate. **(b)** How long does a 60 page report take?



Answer: ________________ minutes

9. Two cookie recipes are shown as tape diagrams of sugar to flour. Which recipe is sweeter, that is, which has more sugar per part of flour? Write each ratio as a fraction of flour, compare them with $>$ or $<$, and name the sweeter recipe.



Answer: ________________

10. Synthesis challenge. A camp cook knows that 15 kilograms of rice feeds 24 people. The double number line below shows a second point: x kilograms feeds 8 people.

(a) Write a proportion from the two pairs and solve it for x. **(b)** Then give the unit rate in kilograms per person.

Answer: ____________________